

PROOF-CARRYING SAT BARRIERS AROUND A 100-VERTEX $R(3, 18)$ NEAR MISS

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ABSTRACT. Let H be a fixed graph on 100 vertices with 827 edges, no independent set of order 18, and the unique triangle $\{97, 98, 99\}$. For a graph F on the same labelled vertex set, write $d_H(F) = |E(H) \setminus E(F)|$; pairs outside $E(H)$ are free and may be added without charge. We give a proof-carrying computation establishing that no triangle-free graph F with $\alpha(F) < 18$ satisfies $d_H(F) \leq 6$. Thus the one-sided deletion repair radius of this fixed seed is at least seven.

The proof first replays a three-branch DRAT certificate excluding deletion budget at most five. At exact budget six, triangle-freeness forces one edge of the unique seed triangle to be absent. Each of the three resulting branches is encoded as a finite CNF relaxation containing all triangle clauses, an exact-five sequential counter for the remaining seed edges, and a checked bank of independent-set hitting clauses. Independent semantic reconstruction and DRAT replay verify all three refutations. We specify the hashes, solver and checker revisions, and the boundary of machine trust. In the exact-seven branch fixing $(97, 99)$ absent, four additional checked DRAT refutations force four named seed edges to be preserved. A checked satisfying model proves that this common relaxation is nonempty but contains an independent 18-set. We further prove that any feasible branch-1 repair has a maximal-triangle-free representative, reconstruct the complete union of four audited independent-set banks, and combine both strengthenings in a byte-audited CNF with 639290 variables and 1972360 clauses. Its one externally bounded 300-second call returned only UNKNOWN; no model, checked refutation, or learned cut resulted. Thus the exact-seven portfolio remains UNKNOWN. In particular, the result is local to H and does not imply $R(3, 18) \geq 101$.

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1. INTRODUCTION

Write $R(3, k)$ for the least n such that every graph on n vertices contains either a triangle or an independent set of order k . The current dynamic survey records

$$100 \leq R(3, 18) \leq 120;$$

see Radziszowski’s continuously maintained table [Rad26]. The public construction of Nagda, Raghavan, and Thakurta supplies a triangle-free graph on 99 vertices with independence number 17, thereby certifying the lower endpoint [NRT26]. Extending that graph by one vertex produces a natural, but logically delicate, local search around the next possible certificate.

This paper studies one labelled 100-vertex extension H . The graph has no independent 18-set but has a unique triangle. It is therefore a near miss, not a Ramsey graph. We ask a deliberately local question: how many *original* edges must be deleted before the triangle can be removed while arbitrary original nonedges are allowed to change into edges? This asymmetric edit model makes additions completely free; consequently, a lower bound in this model is stronger than a lower bound obtained by fixing the nonedges or by restricting the total symmetric-difference distance.

The principal result is a proof-carrying lower bound of seven on that local deletion radius. The argument has four independently inspectable layers.

- (i) A direct graph check establishes the seed identity, its unique triangle, and the absence of an independent 18-set.

- (ii) The unique triangle gives an exhaustive three-way branch split. A previous proof-carrying package excludes total deletion budget at most five, reducing the new layer to exact budget six.
- (iii) Each exact-six branch is translated into a finite CNF *relaxation* of the graph problem. All structural clauses are reconstructed from the seed; the independent-set clauses are validated as universally sound.
- (iv) CaDiCaL emits a DRAT refutation for each branch, and a separately pinned `drat-trim` executable verifies every refutation. Solver status strings are not used as proof evidence.

This layering is important. An unsatisfiable CNF proves the graph claim only after formula soundness and branch completeness have been established. A solver’s report of unsatisfiability proves the CNF claim only after an independently checkable proof trace has been accepted. We record both implications explicitly.

The computation is also informative about what remains open. We first tested the next deletion layer, exact budget seven, with one bounded solver call in each branch. Every call reached its wall-time limit before producing a separated model or a proof. A follow-up branch-1 analysis proves the necessary degree cap $\deg(v) \leq 17$ and thereby excludes 87.5813...% of its raw six-edge residual deletion sets. A deletion-first pilot encoded that lift and initially encountered two apparent exact independent-set separator stalls. An order-aware reconstruction showed that both stalls were vertex-order artefacts: reverse ordering returned the requested witness batches in milliseconds. Seven repaired gates then spent their bounded walls in the master after fast separation, but again produced neither a verified model nor a checked global proof. A subsequent proof-carrying generalized-core route produced four stronger, checked singleton no-goods for branch 1: every satisfying assignment of the common relaxation must preserve each of four named seed edges. Their CNFs were rebuilt byte for byte without the production encoder and their DRAT traces were replayed by separately built checker binaries on two architectures. A checked satisfying model shows that the common relaxation is nonempty, although that model itself contains an independent 18-set. The singleton clauses exclude an additional 1307748575589 residual supports beyond the degree cap, about 2.431110% of its survivors. This is a rigorous family-level pruning result, but it leaves branch 1 and the overall exact-seven problem open. A subsequent exact projection found 396 forced original-nonedge assignments but no additional pure deletion-support clause. We then proved, using the checked budget-six obstruction, that every feasible branch-1 repair has a maximal-triangle-free representative. Combining a full all-pair encoding of that normal form with the complete 530525-mask union gives a fixed CNF with 639290 variables and 1972360 clauses. Independent reconstruction was byte-exact, but its single 300-second bounded call timed out with no model or proof. We record

this and every other unresolved endpoint as UNKNOWN; none is evidence either for or against the existence of a seven-deletion repair.

Claim boundary. The theorem concerns one fixed, labelled seed and one one-sided edit metric. It neither constructs a 100-vertex $(3, 18)$ -Ramsey graph nor rules out such graphs elsewhere in the search space. Thus it does not improve either endpoint of the globally tabulated interval for $R(3, 18)$. The value of the result is instead methodological and local: it gives a complete, proof-carrying obstruction around a particularly close seed, with all negative conclusions separated from incomplete search.

2. THE FROZEN SEED AND THE PRECISE STATEMENT

All vertices are labelled by

$$V = \{0, 1, \dots, 99\}.$$

The input graph H is the adjacency matrix in the artifact

`r3_18_n100_nearmiss.txt`.

Its SHA-256 identity is

`e6527601f72ebb2d3aed11adcdd2fffb0c6aab5e326d81d439c10ee468844e85e`.

Content addressing is part of the mathematical statement: replacing the matrix, even by an isomorphic relabelling, changes the frozen input unless the new identity is separately certified.

Lemma 2.1 (Seed facts). *The graph H has 827 edges, satisfies $\alpha(H) < 18$, and has exactly one triangle, namely*

$$T = \{97, 98, 99\}.$$

Its first 99 vertices induce a graph G_0 with 809 edges, no triangle, and $\alpha(G_0) = 17$.

Computational proof. The matrix parser checks that the input is a symmetric 100×100 binary matrix with zero diagonal. Exhaustive enumeration of the $\binom{100}{3} = 161700$ triples finds only T . A bitset maximum-clique search on the complement returns no clique of order 18, which is equivalent to $\alpha(H) < 18$. The 99×99 principal block agrees entry-for-entry with the separately frozen matrix G_0 , whose SHA-256 is

`3e9d16e29111f3a2f25ae3b992235804c2612de2f6ada5570901d17ceedfca45`.

The same verifier finds no independent 18-set in G_0 and returns an independent 17-set. It also finds no triangle in G_0 . The new vertex has neighbourhood

$$N_H(99) = \{33, 34, \dots, 48, 97, 98\},$$

and the only edge induced by this neighbourhood is $(97, 98)$, providing a second direct check of uniqueness. The negative clique routine was compared with literal subset enumeration on 8640 small random graph/target pairs; all decisions and returned witnesses agreed. Reproduction commands are given in Section 7 and Appendix B. $\square$

Evidence grade and logical role. The assertion $\alpha(H) < 18$ is an exact deterministic trusted-code result: the frozen matrix and inspectable bitset search reproduce it, and exhaustive small-instance comparisons test the pruning invariant. It is not backed by a DRAT/LRAT trace or a proof-assistant certificate. This assertion describes why H is a near miss, but it is not a logical premise of Theorem 2.3. The deletion-barrier proof uses the fixed edge set and seed triangle of H , while the independence-number hypothesis is imposed on the putative final graph F . Thus the implication proved below does not infer anything from $\alpha(H) < 18$.

The graph G_0 reproduces the already tabulated lower bound $R(3, 18) \geq 100$. The graph H , by contrast, contains a triangle and therefore does not certify $R(3, 18) \geq 101$.

Definition 2.2 (Free-addition deletion distance). For a graph F on V , define

$$d_H(F) = |E(H) \setminus E(F)|.$$

Only missing input edges are charged. Every pair in $\binom{V}{2} \setminus E(H)$ may independently be added to F at zero cost. Let

$$\mathcal{F} = \{F \text{ on } V : F \text{ is triangle-free and } \alpha(F) < 18\}$$

and define

$$\rho(H) = \inf\{d_H(F) : F \in \mathcal{F}\},$$

with the convention $\inf \emptyset = \infty$.

Theorem 2.3 (Fixed-seed deletion barrier). *For the content-addressed graph H above,*

$$\rho(H) \geq 7.$$

Equivalently, no $F \in \mathcal{F}$ satisfies $d_H(F) \leq 6$, even when arbitrary subsets of the 4123 original nonedges are added.

The convention in the definition matters: Theorem 2.3 is compatible both with $\rho(H) = 7$ and with a larger or infinite radius. We do not determine which occurs.

3. REDUCTION TO SIX PROOF-CARRYING BRANCH FORMULAS

Let

$$f_0 = (97, 98), \quad f_1 = (97, 99), \quad f_2 = (98, 99)$$

be the three edges of the unique seed triangle. We first state the checked dependency at the preceding deletion layer.

Proposition 3.1 (Budget-five dependency). *There is no $F \in \mathcal{F}$ with $d_H(F) \leq 5$.*

Proof-carrying computation. Every triangle-free F omits at least one f_i . In branch i , the unit $\neg x_{f_i}$ is imposed and a sequential counter permits at most four further deletions among the other 826 input edges. All triangle clauses and a finite bank of universally valid independent-18 hitting clauses are

imposed. The resulting three CNFs have respectively 169382, 184967, and 177799 clauses. All three formulas were reconstructed clause-by-clause from H , and all three DRAT refutations were replayed with a freshly compiled checker from the pinned source revision. The full identities are in Table 4; the formula-soundness argument is the same as Proposition 4.2. Therefore each branch is impossible, and the three branches cover every candidate of total deletion budget at most five. $\square$

By Proposition 3.1, a counterexample to Theorem 2.3 would have exactly six deleted input edges. Fix any one absent edge f_i of the seed triangle. The other five deletions are counted among $E(H) \setminus \{f_i\}$. Notice that the other two triangle edges are not fixed present; if either is also deleted, it simply consumes one of these five residual deletions. Thus the branches cover all candidates and may overlap.

For each $i \in \{0, 1, 2\}$, let Φ_i be the formula defined precisely in Section 4. It contains the fixed negative unit for f_i , an exact-five counter on the residual seed edges, all triangle clauses, and a branch-specific bank of independent-set hitting clauses.

Lemma 3.2 (Exact-six branch cover). *If $F \in \mathcal{F}$ and $d_H(F) = 6$, then the edge assignment of F extends to a satisfying assignment of at least one of Φ_0, Φ_1, Φ_2 .*

Proof. Because F is triangle-free, at least one edge f_i of T is absent. Choose such an i . The chosen unit clause holds. Exactly five members of $E(H) \setminus \{f_i\}$ are absent, so the primary edge assignment extends to the auxiliary variables of the exact-five counter. Triangle-freeness satisfies every triangle clause. Finally, $\alpha(F) < 18$ implies that each installed 18-set contains a present pair, so every installed hitting clause holds. Hence F 's primary assignment extends to a model of Φ_i . $\square$

Proposition 3.3 (Certified branch refutations). *Each formula Φ_0, Φ_1, Φ_2 is unsatisfiable.*

Proof certificate. The three frozen DIMACS files have the identities and dimensions in Table 1. The corresponding DRAT traces were accepted by the pinned `drat-trim` checker, each reporting the exact status line `s VERIFIED`. Before replay, independent semantic checkers reconstructed the entire triangle prefix, cardinality block, fixed unit, and every 153-literal hitting clause; they also checked the compressed and raw identities. Details are given in Section 5 and Appendix B. $\square$

Proof of Theorem 2.3. Suppose $F \in \mathcal{F}$ and $d_H(F) \leq 6$. Proposition 3.1 implies $d_H(F) = 6$. By Lemma 3.2, F 's edge assignment extends to a model of at least one Φ_i . This contradicts Proposition 3.3. Therefore no such F exists and $\rho(H) \geq 7$. $\square$

4. CNF ENCODING AND ITS SOUNDNESS

4.1. Primary variables and structural clauses. For every unordered pair $e = \{u, v\} \in \binom{V}{2}$, a primary variable x_e records whether $e \in E(F)$. The 4950 primary variables are numbered in lexicographic pair order. Crucially, variables exist for all original nonedges; those variables are absent from the deletion counter and are not fixed false.

Triangle-freeness is encoded without separation: for every $a < b < c$, the formula contains

$$\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}.$$

There are exactly $\binom{100}{3} = 161700$ such clauses.

For every installed set $S \in \binom{V}{18}$, the formula contains the positive hitting clause

$$C_S = \bigvee_{e \in \binom{S}{2}} x_e. \quad (4.1)$$

It has $\binom{18}{2} = 153$ distinct primary literals. Every graph with independence number below 18 satisfies C_S . We install only a finite selected bank $\mathcal{B}_i$, not all $\binom{100}{18}$ clauses. Consequently, Φ_i is a relaxation of the exact Ramsey branch. This is the useful direction: unsatisfiability of the relaxation implies unsatisfiability of the exact branch.

4.2. Exact deletion counter. In branch i , let

$$L_i = (\neg x_e : e \in E(H) \setminus \{f_i\})$$

be the ordered list obtained by restricting the lexicographic pair order to the seed edges other than f_i . The list has 826 literals. A PySAT sequential-counter equality encoding implements

$$\sum_{\ell \in L_i} [\ell \text{ is true}] = 5. \quad (4.2)$$

We use the standard sequential counter of Sinz [Sin05]. The concrete generator is exposed by PySAT [IMaMS18]; its semantics is not used as an unchecked black box.

For completeness, define the irredundant sequential schema explicitly. Given signed literals $z_0, \dots, z_{n-1}$ and $1 \leq q \leq n-2$, introduce $y_{r,j}$ for $0 \leq r < q$ and $0 \leq j < n-q$, and let $\text{SC}_{\leq q}(z)$ contain

$$\begin{aligned} \neg z_j \vee y_{0,j} & \quad (0 \leq j < n-q), \\ \neg y_{r,j} \vee y_{r,j+1} & \quad (0 \leq r < q, \ 0 \leq j < n-q-1), \\ \neg z_{j+r+1} \vee \neg y_{r,j} \vee y_{r+1,j} & \quad (0 \leq r < q-1, \ 0 \leq j < n-q), \\ \neg z_{j+q} \vee \neg y_{q-1,j} & \quad (0 \leq j < n-q). \end{aligned}$$

Lemma 4.1 (Sequential-counter projection semantics). *For every assignment of the primary variables,*

$$\exists y \text{ SC}_{\leq q}(z, y) \iff \sum_{j=0}^{n-1} [z_j \text{ is true}] \leq q. \quad (4.3)$$

Consequently, for $n = 826$,

$$\text{SC}_{\leq 821}(\neg z) \wedge \text{SC}_{\leq 5}(z)$$

has an auxiliary extension if and only if exactly five members of z are true. The budget-five block $\text{SC}_{\leq 4}(z)$ has an auxiliary extension if and only if at most four members are true.

Proof. Suppose first that at most q input literals are true. Assign $y_{r,j} = 1$ exactly when at least $r + 1$ of $z_0, \dots, z_{j+r}$ are true. The first clause family records the first threshold crossing. The second propagates an attained threshold to a longer prefix. In the third family, truth of both z_{j+r+1} and $y_{r,j}$ raises the prefix count from at least $r + 1$ to at least $r + 2$. The last family cannot be violated, because such a violation would exhibit $q + 1$ true inputs. This gives the required auxiliary extension.

Conversely, induction on r shows that the clauses force $y_{r,j} = 1$ whenever at least $r + 1$ of $z_0, \dots, z_{j+r}$ are true. For $r = 0$, use the first clause at the position of a true input and then the propagation clauses. For the induction step, let p be the position of the $(r + 1)$ -st true input and put $j' = p - r$. The induction hypothesis forces $y_{r-1,j'}$; the third family, together with z_p , forces $y_{r,j'}$, and propagation reaches $y_{r,j}$. If $q + 1$ inputs were true, let p be the position of the $(q + 1)$ -st and put $j = p - q$. The preceding argument forces $y_{q-1,j}$, while the last family forbids its conjunction with z_p , a contradiction. This proves (4.3). Applying (4.3) to z and to its 826 complements gives respectively the upper and lower sides of the equality claim; the at-most-four claim is the direct case $q = 4$. $\square$

In the exact-five DIMACS block, the complement counter at bound 821 comes first and has 4105 auxiliary variables and 7394 clauses; the positive counter at bound 5 has another 4105 auxiliaries and 9026 clauses. Thus the equality block has 8210 auxiliary variables and 16420 clauses, ending at variable 13160. The checker `routes/finite/check_r3_18_seqcounter.py` does not import or call PySAT's cardinality generator. It independently generates the four clause families, compares all six actual counter blocks clause-for-clause in their canonical literal order, checks their digests and variable ranges, and constructs the prefix-threshold auxiliary assignment from the proof above. Exhaustive small-instance tests enumerate both primary and auxiliary assignments and verify both directions of the projected at-most and equality semantics.

Thus

$$\Phi_i = \left(\bigwedge_{a < b < c} (\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}) \right) \wedge \text{Card}_{=5}(L_i) \wedge \neg x_{f_i} \wedge \bigwedge_{S \in \mathcal{B}_i} C_S. \quad (4.4)$$

TABLE 2. Audited construction of the branch-0 universal cut union.

source	source masks	new masks	cumulative
branch-0 discovery checkpoint	183133	183133	183133
branch-0 earlier bank	141149	4096	187229
branch-1 bank	63943	52515	239744
branch-2 bank	5422	3931	243675
budget-five branch 0	287	286	243961
budget-five branch 1	15872	4262	248223
budget-five branch 2	8704	3548	251771

TABLE 1. Exact-six branch dimensions. All branches have 4950 primary variables, 8210 counter auxiliaries, 161700 triangle clauses, 16420 counter clauses, and one fixed unit.

branch	fixed absent edge	hitting clauses	variables	total clauses
0	(97, 98)	251771	13160	429892
1	(97, 99)	63943	13160	242064
2	(98, 99)	5422	13160	183543

Proposition 4.2 (CNF soundness). *For each i , every graph $F \in \mathcal{F}$ with $d_H(F) = 6$ and $f_i \notin E(F)$ induces a primary assignment that extends to a model of Φ_i .*

Proof. The triangle block holds by triangle-freeness. The fixed unit holds by the branch premise. Since f_i is one of exactly six missing input edges, exactly five literals in L_i are true; Lemma 4.1 supplies an auxiliary assignment. Each $S \in \mathcal{B}_i$ contains a present pair because $\alpha(F) < 18$, so every C_S holds. No condition was imposed on an original nonedge except through triangle and hitting clauses, hence arbitrary additions remain represented. $\square$

4.3. Sound reuse of the branch-0 cut union. Branch 0 uses 251771 distinct 18-sets deduplicated from seven checked sources. The ordered mask digest is

`f10690b826b86eb03567a2ffa2fb553801fae32af20cfe4337118bddf4e41afa.`

Only vertex-set masks were transferred. No learned clause, branch assumption, or solver conclusion was reused. Since every clause C_S in (4.1) follows solely from $\alpha(F) < 18$, it is valid in every branch. The source counts and cumulative deduplication totals appear in Table 2.

TABLE 3. Exact-six proof artifacts and replay scale. Hashes are abbreviated here and given in full in Table 5.

branch	raw DRAT bytes	proof lines	compressed bytes	replay
0	3493847713	6649939	656144450	verified
1	2137671968	5590150	433552903	verified
2	759008359	2837771	152784369	verified

5. PROOF PRODUCTION, CHECKING, AND TRUST

5.1. From finite formulas to checked refutations. The final proof-producing runs used CaDiCaL [BFF⁺24] at source commit

c60730422e758ef1cebe7aeddff2dda31c996bf04.

An exit code 20 and the text `UNSATISFIABLE` were treated only as requests to check the emitted proofs. The final mathematical evidence is the successful replay of the DRAT traces [WHH14, Heu16] by the official `drat-trim` source at commit

2e3b2dc0ecf938adbbd779d42877b6ed69d9a985.

The principal recorded executable has SHA-256

31d9e8e04bc76a65f0a18ea212ac44af4cf7bc398629495cec093ced772c50f4.

All three checks terminated with exit code zero and reported an exact `s VERIFIED` status line.

For branch 0, a fresh replay in a temporary directory took 273.448 seconds inside the checker and reported 135152646 resolution steps in the retained core. Branch 1 reported 209897525 resolution steps in its independently checked run. These timings are descriptive, not logical premises. The source commit and successful replay are the portable evidence; the executable hash records the historical build and is not required to match a fresh binary compiled on another platform.

5.2. Semantic reconstruction. The proof checker establishes unsatisfiability of the bytes in each DIMACS file. Separate semantic checkers establish that those bytes are the intended relaxations. For every branch they independently:

- (1) parse the frozen seed and regenerate the lexicographic pair-variable map;
- (2) compare all 161700 triangle clauses in order;
- (3) regenerate the production PySAT counter and, in a separate checker that does not import `CardEnc`, independently generate the sequential-counter schema and compare its variables and clauses in order;
- (4) check the branch unit and the exact clause partition;
- (5) reconstruct every hitting clause as all 153 pair variables of one 18-vertex mask;

- (6) verify the compressed and raw sizes and SHA-256 identities; and
- (7) verify the budget-five dependency and the three-edge cover.

Branch 0 additionally reparses all seven cut sources and reproduces the deduplicated union. Branches 1 and 2 likewise reconstruct their complete stored cut banks. An earlier interrupted branch-1 trace fails gzip integrity, remains quarantined, and is excluded from every manifest and conclusion.

5.3. Trust boundary. The claim is proof-carrying, not formally verified in a proof assistant. Its earliest machine dependencies are:

- the bitset independent-set verifier used for the descriptive seed fact $\alpha(H) < 18$, which is not a premise of the deletion-barrier implication;
- the `drat-trim` proof kernel, together with its compiler and execution hardware; and
- the independently inspectable scripts that map matrices, sequential schemas, and masks to DIMACS.

The audit narrows this boundary through independent matrix parsing, exhaustive triangle enumeration, small-instance cross-checks of the clique routine, exhaustive small-instance projection tests for the independent counter schema, constructive auxiliary assignments, complete clause reconstruction, content hashes, gzip integrity tests, and proof replay with a freshly compiled checker. PySAT remains production provenance, but Lemma 4.1 and the `CardEnc`-free comparison mean that its counter semantics is no longer assumed merely because PySAT emitted the bytes. A future LRAT conversion and formally verified LRAT checker could reduce the trusted kernel further, but is not required for the conventional proof-carrying SAT standard used here.

6. THE UNRESOLVED EXACT-SEVEN LAYER

The theorem implies only $\rho(H) \geq 7$. To distinguish this lower bound from an exact value, we report bounded experiments at deletion budget seven.

In branch i , the fixed unit $\neg x_{f_i}$ accounts for one deletion and an equality counter requires exactly six of the other 826 input edges to be absent. All 4123 original nonedges remain free. The universal bank of 251771 hitting clauses from branch 0 is valid in every branch and was preloaded. Every initial formula therefore had 4950 primary variables, 9840 counter auxiliaries, 161700 triangle clauses, 19680 counter clauses, one fixed unit, and 251771 hitting clauses, for a total of 433152 clauses.

Computational observation 6.1 (Bounded endpoints, not a theorem). One solver call was launched per branch with a parent-enforced 300-second discovery wall. Branch 0 used CaDiCaL, branch 1 Glucose 4.2, and branch 2 MapleChrono. Each remained inside its first SAT call until the wall expired. No branch produced a model that reached independent-set separation, and no branch produced a checked refutation. The endpoint of all three branches is therefore UNKNOWN.

The zero recorded separation iterations means that no SAT model returned before interruption; it does not mean the formula is unsatisfiable. The recoverable state consists of the completely reconstructed initial formula and universal cut bank. Internal CDCL state was not serialized. Exact hashes of the frozen result and checkpoint records are listed in Appendix C.

Proposition 6.2 (A degree-cap lift for branch 1). *Let F be triangle-free with $\alpha(F) < 18$. Then $\deg_F(v) \leq 17$ for every vertex v . In branch 1, where (97, 99) is fixed absent, every feasible set of six residual seed-edge deletions contains an edge incident with vertex 98. Consequently this condition alone excludes*

$$\frac{\binom{808}{6}}{\binom{826}{6}} = 0.875813014052041 \dots$$

of the raw residual deletion sets.

Proof. The open neighbourhood of any vertex in a triangle-free graph is independent, so its size is at most 17. After deleting the fixed branch edge (97, 99), vertex 98 still has seed degree 18. If D denotes the six residual deletions and A the selected seed-nonedge additions, then

$$\deg_F(98) = 18 - |D \cap \delta(98)| + |A \cap \delta(98)| \leq 17.$$

Thus D must meet the 18 residual seed edges incident with vertex 98. There are 808 residual edges not incident with 98, so precisely $\binom{808}{6}$ of the $\binom{826}{6}$ raw six-edge sets are excluded. $\square$

For the rest of this section put

$$E_1 = E(H) \setminus \{(97, 99)\}, \quad \Delta_{98} = \{e \in E_1 : 98 \in e\}.$$

Thus $|E_1| = 826$, $|\Delta_{98}| = 18$, and a branch-1 repair has a residual deletion support $D \in \binom{E_1}{6}$.

Lemma 6.3 (Soundness of the branch-1 common relaxation). *Let Ψ_1 be the CNF consisting, in order, of all triangle clauses, an exact-six counter on the literals $\{\neg x_e : e \in E_1\}$, the unit $\neg x_{97,99}$, one degree-at-most-17 counter for every vertex, and the ordered bank of 251771 universal independent-18 hitting clauses. Every feasible branch-1 exact-seven repair extends to a satisfying assignment of Ψ_1 .*

Proof. The primary variables x_{uv} record the final graph. Triangle-freeness satisfies every triangle clause and the branch choice satisfies the fixed unit. Exactly six edges of E_1 are absent, so the primary assignment has an extension through the exact counter. The degree-cap proposition gives an extension through every degree counter. Finally, for each stored 18-set S , the clause

$$\bigvee_{\{u,v\} \in \binom{S}{2}} x_{uv}$$

is necessary when $\alpha(F) < 18$. Hence every clause family is satisfied. The bank need not contain every 18-set, so Ψ_1 is a relaxation rather than an exact encoding. $\square$

Proposition 6.4 (Four proof-checked singleton entailments). *Define*

$$a = (11, 62), \quad b = (18, 61), \quad c = (18, 64), \quad d = (18, 69).$$

All four edges lie in $E_1 \setminus \Delta_{98}$. The frozen relaxation satisfies

$$\Psi_1 \models x_a \wedge x_b \wedge x_c \wedge x_d.$$

Thus none of these four seed edges can belong to the residual deletion support of a feasible branch-1 repair. No minimality assertion is made.

Proof. For each $e \in \{a, b, c, d\}$, a standalone CaDiCaL run refuted

$$\Psi_1 \wedge \neg x_e,$$

and the pinned `drat-trim` checker accepted each emitted proof. These four checked refutations are equivalent to the displayed entailment by Ψ_1 . They are not asserted to be equivalent to the target-graph statement; the necessary condition for target graphs follows only in the forward direction from Lemma 6.3.

Each DIMACS instance has 154190 variables and 718453 clauses. A PySAT-free checker that imports neither the production formula builder nor the exporter reconstructed all four compressed CNFs byte for byte, including the candidate-unit polarity. A separately compiled checker binary on a second architecture replayed all four proofs to the exact status `s VERIFIED`. The formula, proof, checker, and artifact identities are recorded in Appendix C. $\square$

Computational observation 6.5 (The common relaxation is nonempty). A complete CaDiCaL model for Ψ_1 was checked against an independent byte-for-byte reconstruction of all 718452 clauses. The model sets all four displayed edge variables to true, as the proposition requires. It also contains the independent 18-set

$$\{99, 96, 97, 95, 92, 90, 87, 91, 85, 93, 84, 73, 70, 88, 82, 81, 78, 83\}.$$

Hence the singleton entailments are not consequences of an empty common relaxation, but this particular model is not an exact-seven Ramsey repair. The satisfiability of Ψ_1 resolves neither existence nor nonexistence for the target problem because the universal bank is incomplete.

Corollary 6.6 (Support exclusion beyond the degree cap). *Every feasible branch-1 residual support D meets Δ_{98} and avoids $\{a, b, c, d\}$. The singleton no-goods exclude an additional*

$$1307748575589$$

supports beyond those already excluded by the degree cap. Together the two filters exclude

$$380670699850897$$

of the $\binom{826}{6} = 433155188594590$ raw supports and leave 52484488743693 outside these two necessary-condition filters.

Proof. The support restrictions follow from the degree-cap proposition, Lemma 6.3, and Proposition 6.4. Avoiding the four distinguished edges leaves 822 eligible residual edges. Because those four edges avoid Δ_{98} , the supports that both avoid them and fail the degree test are precisely the six-subsets of the other 804 nonincident edges. Hence the number surviving both filters is

$$\binom{822}{6} - \binom{804}{6} = 52484488743693.$$

Subtracting this from the degree-cap survivor count gives the additional 1307748575589 exclusions. Equivalently, the singleton clauses remove about 2.431110% of the degree-cap survivors, and the combined filters exclude about 87.883214% of the raw support universe. These are support-index counts, not a wall-time speedup or a count of graphs. $\square$

The four retention units also permit a stricter audit of what follows from the local triangle and degree systems before any new SAT call. Propagating those systems to the original nonedges gives exactly 396 forced-absent addition variables: 388 come from pairs having more than six common neighbours in the fixed branch base, two are forced by the retained-star configuration, and six arise only from the coupled exact-six, $\delta(98)$, and degree constraints. Substitution of all 396 units into the finite I_{18} bank leaves no clause involving deletion variables alone. Thus this closure is useful preprocessing, but it supplies no new deletion-support no-good beyond the filters proved above.

There is, however, a clean normal form that applies to the target problem rather than merely to a finite bank.

Proposition 6.7 (Maximal-triangle-free normal form). *A feasible branch-1 exact-seven repair exists if and only if one exists that is maximal triangle-free. Equivalently, one may require every absent pair to have a common neighbour.*

Proof. Only the forward direction needs proof. Starting from a feasible repair F , repeatedly add an absent pair $uv \notin E(H)$ whenever u and v have no common neighbour. Such an addition creates no triangle, cannot increase the independence number, and does not change $E(H) \setminus E(F)$. The finite process terminates with the same exact seven-edge deletion support and with every still-absent original nonedge triangle-blocked.

It remains to consider an absent pair $e \in E(H)$. If e were safe to add, doing so would again preserve triangle-freeness and could not increase the independence number, while reducing the one-sided deletion count from seven to six. This contradicts the proof-checked theorem $\rho(H) \geq 7$. The argument also applies when $e = (97, 99)$: after adding that fixed branch edge, at least one of the other two edges of the seed's unique triangle remains absent, so the resulting graph is covered by the global three-branch budget-six exclusion. Hence every nonedge in the final graph has a common neighbour,

which is precisely maximal triangle-freeness. The reverse implication is immediate. $\square$

This is an existence normal form, not a claim that every raw repair is already maximal. It is the fixed-support analogue of the standard use of maximal triangle-free graphs in computational Ramsey searches [BGSP12]. For a scoped pair uv , an exact existential CNF encoding introduces variables $y_{uv,w}$ and uses

$$x_{uv} \vee \bigvee_{w \notin \{u,v\}} y_{uv,w}, \quad \neg y_{uv,w} \vee x_{uw}, \quad \neg y_{uv,w} \vee x_{vw}.$$

Reverse implications are unnecessary: the y -variables only select one common-neighbour witness when uv is absent. Scoping this construction to the 4123 original nonedges uses 404054 auxiliaries and 812231 clauses; the full maximal-triangle-free encoding uses 485100 auxiliaries and 975150 clauses. A complete three-vertex truth-table audit checks the existential semantics.

The two checked relaxation models illustrate the claim boundary. The common model has one unsafe original nonedge, (37, 86). Adding it would create a degree-18 vertex, so the resulting graph is not asserted to remain a model of the degree-capped incomplete-bank relaxation; the same pair instead recovers the independent-18 domination witness recorded below. The A+ model has exactly three unsafe original nonedges, (16, 64), (32, 98), (40, 89). Adding all three gives a maximal triangle-free primary graph with the same seven deletions and maximum degree 17, but it still contains the already known independent set $\{81, 82, \dots, 97, 99\}$. Maximality is therefore complementary to, not a replacement for, independent-set separation.

Nor can a branch-1 conclusion be transferred by relabelling. A deterministic degree-initialized one-dimensional Weisfeiler–Leman refinement has 3, 15, and 100 color classes at initialization, round one, and round two. Every automorphism preserves every refinement color, so the singleton round forces the automorphism group of the labelled seed to be trivial. The other two triangle-edge branches therefore require independent treatment.

For completeness, the domination diagnostic used above is the following elementary consequence of the target conditions. Put

$$Z_v = \{u \neq v : uv \notin E(F), N_F(u) \cap N_F(v) = \emptyset\}.$$

Then $N_F(v) \cup I$ is independent for every independent $I \subseteq F[Z_v]$, and hence $\alpha(F[Z_v]) \leq 17 - \deg_F(v)$. In particular, degree 17 forces $Z_v = \emptyset$, while degree 16 forces Z_v to be a clique of order at most two. On the common model this exposes the new mask 2000000404414192982822020; it has zero overlap with the universal, historical, A+, and fixed-base mask families. On the A+ model it yields no new mask, consistently with all three unsafe pairs having degree 16 at both endpoints. These are checked separators for relaxation models, not new theorems about existence.

We encoded the same degree cap at both levels of a deletion-first branch-1 pilot. The master selects six residual deletions and addition-eligibility variables, separates conditional independent-set clauses, and passes a fixed deletion set to an exact add-only subproblem. An incomplete master separation call may fall back to that exact subproblem. If the fixed-deletion subproblem is itself incomplete, the run stops without learning a deletion no-good. A completed fixed-deletion UNSAT may be used as solver-trusted search telemetry, but it is not promoted without a checked proof trace.

Computational observation 6.8 (Deletion-first endpoints, still not a theorem). A six-configuration bounded matrix learned between zero and 2816 new conditional independent-set cuts per run. One MapleChrono configuration reached two fixed-deletion UNSAT endpoints, but emitted no proof for them; the same run then stopped at the independent-set oracle limit. Three predeclared positive-signal extensions raised that oracle limit from two million to twenty million nodes. Two stopped at exactly 20,000,001 recursive nodes inside a fixed-deletion subproblem after learning 10,780 and 13,362 new cuts, respectively; the third learned 12,800 cuts but reached its 300-second global wall before entering a subproblem. No run returned a verified candidate or a checked refutation. Every endpoint therefore remains UNKNOWN.

The two apparent fixed-deletion separator stalls were then reconstructed under an order-aware implementation. The historical ascending recursion exhausted its two-million-node allowance on both supports. Reversing the vertex order, which leaves the graph property unchanged, instead returned 512 validated independent 18-sets in 11,261 and 10,057 recursive nodes, respectively, taking about 0.006 seconds in each case. Thus the old twenty-million-node endpoints were an ordering artefact on these supports, not evidence of a combinatorial obstruction.

We consequently ran seven further bounded gates with a reverse-first, bidirectional exact oracle. On every master model for which separation was invoked, a batch of either 512 or 4096 validated cuts was returned in 0.005–0.075 seconds. Nevertheless, each gate reached its global wall after zero to five master models; none entered a fixed-deletion subproblem, returned a verified candidate, emitted a proof, or learned a new fixed-deletion no-good. Under these particular checkpoints and limits, every run reached its global wall while still in the master. This remains bounded, configuration-specific telemetry rather than a claim of intrinsic hardness. The command lines, stable implementation hashes, all sixteen deletion-first endpoints, and the stop rule are recorded in Appendix C.

We also tested a genuinely different deletion-only projection. It retains the exact residual counter, local wedge implications for addition eligibility, the complete 235504-member fixed-base I_{18} covering family, and the vertex-98 hit constraint, while forgetting the full edge assignment. Its natural exact-six control has 13159 variables and 264757 clauses. A pinned CaDiCaL call reached its predeclared 300-second wall without a status line. The

incomplete proof prefix was hashed and deleted, and the exact-seven instance was not run under the predeclared stop rule. This endpoint is UNKNOWN; more importantly, the route failed its strength gate on an already known-unsatisfiable control and therefore supplied no reason to spend a second long call.

Finally, a no-repeat A+ bank was generated from a checked common-relaxation model after excluding the universal, historical, and exhaustive fixed-base families. Appending its 4096 new I_{18} clauses and the four proved retention units produced a formula with 154190 variables and 722552 clauses. The single authorized solver call returned a complete satisfying model after 2532.362 seconds. An independent checker evaluated every clause, recovered a triangle-free 823-edge primary graph, and found the independent 18-set

$$\{81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 99\}.$$

That set was absent from the installed universal and A+ banks but already belonged to both the historical and exhaustive fixed-base families. Thus A+ is a rigorously audited negative endpoint and, specifically, shows why a fresh prefix is inferior to installing the complete deduplicated bank. It does not provide a repair or a branch refutation.

We therefore reconstructed that stronger bank explicitly. The universal, historical, and exhaustive fixed-base families contain 251771, 64591, and 235504 masks, respectively; their deduplicated union has 526429 members. The 4096 A+ masks are disjoint from all three, giving 530525 distinct necessary I_{18} clauses. Because the common formula already contains the universal family, the strengthened formula appends 278754 new hitting clauses and the four retention units. Its deterministic DIMACS image has 154190 variables, 997210 clauses, 390604816 bytes, and SHA-256

4f7e8f5b724a657888c7814d2c25cac41c283c3db356fb8f1a15f4c7322c375d.

An independent reconstruction agrees byte for byte. This is strictly stronger than A+: it installs the two families that were previously used only as no-repeat filters and directly blocks the displayed A+ witness. It also differs from the deletion-first master with all fixed-base cuts, because it retains the full edge formula and adds 43614 history-only masks plus the 4096 A+ masks. Formula construction alone is not a SAT or UNSAT result.

We finally combined this complete mask bank with the full all-pair maximal-triangle-free encoding of Proposition 6.7. The resulting direct primary-graph CNF has 639290 variables, 1972360 clauses, 408370088 bytes, and SHA-256

09e1784c3f43c4901dc6f6b4749fc5a74b025b08f74be58133edd1ed1096ebdb.

A separate reconstruction agreed byte for byte, both before and after a command-line symlink-validation repair that did not change the formula. Exactly one externally bounded Sirius call used CaDiCaL 3.0.1 at source commit c60730422e758ef1cebe7aeddff2dda31c996bf04, a 300-second wall, non-binary DRAT, and a 10-GiB proof cap. The wrapper exited 124, and the result file contained exactly c UNKNOWN. The run recorded 13347 conflicts,

105834 decisions, and 1216037481 propagations; these are resource telemetry only. Its incomplete 922038097-byte proof prefix was hashed and deleted without replay. Consequently the run produced no model, certificate, learned cut, branch closure, or global Ramsey implication.

Accordingly, none of the following is established: a repair with exactly seven deletions; nonexistence of such a repair; the exact value of $\rho(H)$; or a 100-vertex (3, 18)-Ramsey graph. A machine-readable Boolean field whose default value is false is likewise not evidence of nonexistence unless the companion establishment flag and a checked proof support it.

7. REPRODUCIBILITY AND ARTIFACT PUBLICATION

7.1. Two verification tiers. The public source repository [Xin26] separates lightweight tracked checks from large proof replay. Its accompanying evidence release is designated at <https://github.com/jackal092927/ramsey-bounds-reproducible/releases/tag/evidence-2026-08-30> and is valid only if it contains exactly the 26 assets listed in `artifacts/MANIFEST.tsv`. Publication, immutability, credential-free download, and final-tag replay are release gates recorded separately; they are not premises of the graph theorem. Tier 1: source and semantics. Python 3.11.15 and the locked environment reproduce graph verification, unit tests, the PySAT-independent sequential-counter schemas and their frozen block digests, and exact-seven frozen-state checks. From the repository root, the lightweight entry point is

```
bash scripts/bootstrap.sh
.venv/bin/python reproduce.py quick
```

The retained maximal-union timeout endpoint is audited separately, without recreating or promoting its deleted proof prefix, by

```
.venv/bin/python \
  routes/finite/check_r3_18_budget7_branch1_maximal_union_probe.py
```

That checker requires exit 124, the exact `c UNKNOWN` result, the fixed formula and solver hashes, and the absence of any retained file named as a proof certificate. The proof-critical Python packages are pinned to `python-flint==0.9.0`, `mpmath==1.4.1`, `python-sat==1.9.dev13`, and `six==1.17.0`. The finite route uses PySAT; the other entries are repository-wide proof dependencies.

Tier 2: large proof replay. Once published, the release is downloaded without GitHub credentials by

```
bash scripts/download_release_artifacts.sh
```

The helper first requires the complete release asset-name set to equal the manifest exactly and then checks the content before decompression:

```
.venv/bin/python scripts/verify_artifacts.py \
  --directory /path/to/release-assets --exact
```

The exact mode also rejects unmanifested files, so a stale or mixed download directory cannot masquerade as the immutable release payload. The manifest includes audited Linux x86-64 and macOS arm64 `drat-trim` binaries, their source-commit marker, and the upstream license. The strict

branch-1 replay accepts exactly either audited binary hash; `scripts/build_drat_trim.sh` remains a non-normative manual comparison helper. The one integrated command

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir /path/to/release-assets \
  --drat-trim /absolute/path/to/audited-manifest-binary
```

materializes a temporary overlay containing tracked semantic metadata and the 26 release assets. It compares the independent sequential schemas with all six theorem counter blocks, reconstructs the six theorem formulas and four singleton formulas, replays all ten proofs, checks the complete common relaxation model, and audits the exact-seven state. A proof is accepted only when the checker exits zero and prints an exact `s VERIFIED` status line.

7.2. Why large proofs are release assets. The six exact-six compressed files already occupy about 1.26 GB, and the four singleton compressed proofs add several more gigabytes. Adding these or the smaller budget-five proofs to ordinary Git history would make cloning needlessly expensive. The repository therefore tracks code, matrices, cut banks, semantic records, tests, the paper, and a content-addressed manifest. CNF and DRAT payloads are designated for distribution as immutable GitHub Release assets, keyed by SHA-256 and byte count. No DOI-bearing archival mirror is claimed. This design keeps ordinary clones reviewable while preserving bit-for-bit replay.

The release is complete only if it includes the three budget-five proof pairs used by Proposition 3.1, the three exact-six proof pairs used by Proposition 3.3, the four strongest singleton proof pairs, the common CNF/model pair, and the audited checker toolchain files. Their identities are listed in Tables 4, 5 and 9 and in the strict manifest. Dominated pair-core, incomplete, corrupt, and exploratory traces are explicitly excluded.

7.3. Reproducibility contract. A reproduction should report five logically distinct outcomes:

- (1) the seed and artifact hashes match;
- (2) semantic reconstruction passes;
- (3) each required DRAT replay is verified;
- (4) the common-relaxation SAT model is checked and its independent 18-set is recovered; and
- (5) the exact-seven endpoint remains UNKNOWN, unless a new, separately reviewed certificate supersedes it.

Passing only a solver run, only a hash check, or only unit tests does not by itself reproduce the full theorem.

8. SCOPE, LIMITATIONS, AND CONCLUSION

We established a complete local statement: every triangle-free graph with independence number below 18 on the labelled vertex set of H deletes at least seven of H 's 827 edges, even if additions among all original nonedges

are free. The proof is carried by six checked SAT refutations: three for budget at most five and three for the exact-six layer.

The conclusion does not transfer automatically to an isomorphism class, to a different extension neighbourhood, or to a symmetric edit metric. It does not rule out a remote 100-vertex construction, and it does not improve the global bound on $R(3, 18)$. The exact-seven layer remains open. These are mathematical scope restrictions, not merely caveats about running time.

The main methodological lesson is that a computational near miss can support a rigorous negative result without being oversold as a Ramsey-number improvement. A content-addressed seed, a proof of case coverage, independently reconstructed CNF semantics, and replayable UNSAT proofs together establish a precise local barrier. Keeping bounded UNKNOWN runs outside the theorem makes the remaining target equally precise: either exhibit a seven-deletion repair and verify it directly, or produce proof-carrying refutations of all exact-seven branches.

APPENDIX A. CONTENT-ADDRESSED ARTIFACT LEDGER

The digests in the two artifact tables below are SHA-256 identities of the compressed files. Their byte counts refer to those same compressed files and therefore permit a check before decompression. Digests explicitly introduced as uncompressed identities or as metadata-record identities have the separate meaning stated in their surrounding text.

Table 4: Budget-five proof pairs.

branch	artifact	bytes	SHA-256
0	CNF	837880	d64f009f840597e266ccfb3b31c88f7e87eb760 d1fc551b7ee08a0854749ba66
0	DRAT	13125926	60687241833df8ae957b3b4fcb02e6e9f7595c6 e3dcbee8b77f02dad5443f913
1	CNF	1797965	d6d321a4ac1aee5e99c48843a8cda9395271e8 9555cea328e29bcda4e70ef15
1	DRAT	37102450	33a9e037e19beeab8dd2af4dca5ce9010d4906b 39dc3e5cc8fd62823f518497a
2	CNF	1363549	8d8a543bb3cc17014941fee432673a1dc2b99c8 cb4885d4d91bba9005f3b10a7
2	DRAT	21678734	159865da9ad23eea298ebd4e9406f2f37ecc601 5a593975e9d63e3b08a859897

The corresponding uncompressed identities, independently recomputed during review, are as follows:

Branch 0 CNF: c907dc96c1ff3378d345ead0ea22456eccef50f321cc53b6cf7dda46784d6389

Branch 0 DRAT: 517d076efd026eab6eeb89583c0226e713be968120344e6b82c3471
e0efe6b5d
Branch 1 CNF: 5ba541f58ea42d7fb73ff51a962e98153eece0d4a8bf13c1bea3a86ed
0d50429
Branch 1 DRAT: 84a2236b7692207f59c3d2dcb3521b2f2b947428470384ac643c34a
7bfbf69
Branch 2 CNF: 441b70e023f4a9fa1837f16b1502ef3b54a2f00b6143fd5e7de0131ac
95beebf
Branch 2 DRAT: 4ff3a43bda7774c09695417a9f9684e8a856fb46bf62225a08e746f
bf37d1ad6

Table 5: Exact-six proof pairs.

branch	artifact	bytes	SHA-256
0	CNF	17600735	47052de808b98598f2f144251a6ad73c1415dd0 0fef9e04b99ffe2f176229fbf
0	DRAT	656144450	4d948ab34f1d2475af6efe943e0a4899827c78b 96c2bc3835b5161f189c26bb5
1	CNF	5052085	a438c9fef4d99ae516829e4405da8c9f0793985 01361f9c5f69f46a494ffad14
1	DRAT	433552903	cfe694fd728903ee9ac9f08a66ade64faf7e687 63246d1d52ca29a85727494ec
2	CNF	1205208	c51fed43eeca3becf64bcdfe747a4fdf3e516c e0e6a02d7d867941b87a99c06
2	DRAT	152784369	c9648a4e38c17dcdfe97d873095eda002abc689 b3b3bd93f02fc8a369198c4b9

For the exact-six formulas, the frozen cut-bank identities are

Branch 0, 251771 masks: 91b5709248ff641a315f5a0389b4f3fde3d38514f3b1a8
b31b6cad31224f250b
Branch 1, 63943 masks: f8064f0f226659b4750382f781ba9a862bcc19eb49006eb
8cdd2afc002ed227c
Branch 2, 5422 masks: aaa79a2f25d4c0d61c4467f1d0a721fd670a624e0391d96b
2122723f97ba4598

The exact-six proof records have SHA-256 identities

Branch 0: 20729b2a818aaca35cdee6e9deeb5a0e
6e4bbd6dc5ca11e5abb6dfe9409b035f
Branch 1: 4a3e7b4c729f696ff304c5f71621e018
786b552146784a2141529a4c6f028666
Branch 2: ee0c73b80407a4dd48ca69bd1a666f39
1bfd5729b9a56e062deef2a6f62018de

The source files use the following release-asset names:

r3_18_budget5_branch_0_twostage.{cnf,drat}.gz
r3_18_budget5_branch_1.{cnf,drat}.gz

```

r3_18_budget5_branch_2.{cnf,drat}.gz
r3_18_budget6_branch_0_universal_union.{cnf,drat}.gz
r3_18_budget6_branch_1.{cnf,drat}.gz
r3_18_budget6_branch_2.{cnf,drat}.gz

```

APPENDIX B. INDEPENDENT VERIFICATION PROTOCOL

This appendix states a reviewer-oriented protocol. It is intentionally more specific than the convenience wrapper in the repository.

B.1. Seed verification. Parse the 100×100 matrix, reject any non-binary, asymmetric, or non-zero-diagonal input, and verify its SHA-256. Count undirected edges by the strict upper triangle. Enumerate every vertex triple and record all triangles. Run the exact independent-set verifier at target 18. Repeat the structural checks for the 99-vertex principal submatrix and compare it byte-for-byte with its separately stored certificate. A successful outcome must report 827 edges and exactly the triangle $\{97, 98, 99\}$ for H , and 809 edges, no triangle, and no independent 18-set for G_0 .

B.2. Formula reconstruction. For each exact-six branch i :

1. Assign primary identifiers $1, \dots, 4950$ to pairs in lexicographic order.
2. Generate the 161700 triangle clauses in lexicographic triple order and compare them to the DIMACS prefix.
3. Form the 826 residual deletion literals in the lexicographic order inherited from the primary pair list. Without calling PySAT’s `CardEnc`, generate the four Sinz clause families in Lemma 4.1. Generate the equality block as at-most 821 on the complemented literals followed by at-most 5 on the original literals. Compare all 16420 clauses and the final variable identifier 13160.
4. Compare the fixed negative unit for f_i .
5. For each remaining clause, require 153 distinct, positive primary literals. Invert the pair map, recover exactly 18 vertices, and verify that the clause is the full $\binom{18}{2}$ pair family of those vertices.
6. Compare the recovered ordered masks with the frozen cut bank and check that no unparsed clause remains.

The analogous budget-five reconstruction replaces equality at five by an at-most-four counter. It has 8238 total variables and 7394 counter clauses in each branch. The three hitting-bank sizes are 287, 15872, and 8704. The executable audit is `routes/finite/check_r3_18_seqcounter.py`; its schema-only mode is part of the lightweight tier, and the complete mode compares the six actual DIMACS blocks during `finite-heavy`.

For each branch-1 singleton instance, independently reconstruct the common 718452-clause relaxation including all degree counters, append exactly one negative primary-variable unit, and require 154190 variables and 718453 clauses. The four units must be $-1085, -1672, -1675, -1680$. Separately reconstruct the unit-free common CNF, parse a complete assignment of all

154190 variables, evaluate every clause, and run an exact independent-set verifier on the recovered graph.

B.3. DRAT replay. Use one of the two manifest-authenticated `drat-trim` binaries built from source commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. Verify the adjacent source marker and the exact platform-binary hash. The repository helper

```
bash scripts/build_drat_trim.sh \
  /absolute/new/checker-directory
```

performs a source checkout for non-normative manual comparison; an arbitrary fresh build is not accepted by the strict core-proof gate. For each of the ten proof pairs in Tables 4, 5 and 9, check compressed byte counts and SHA-256 values, test gzip integrity, and decompress into a new temporary directory. Then invoke

```
/path/to/drat-trim formula.cnf proof.drat
```

and require both exit status zero and an exact output line `s VERIFIED`. The checker may print timing diagnostics after that status line. Delete the temporary raw files only after recording the uncompressed hashes.

For the complete package, the repository offers integrated semantic and proof replay:

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir /path/to/release-assets \
  --drat-trim /path/to/audited-manifest-binary
```

Time limits may need to be increased on slower hardware. A timeout is UNKNOWN, never a failed or successful proof.

B.4. Logical audit checklist. A reviewer can check the mathematical implication independently of the code:

1. the three f_i are precisely the edges of a seed triangle;
2. every triangle-free final graph omits at least one f_i ;
3. original nonedges are primary variables but not deletion literals;
4. every installed C_S is entailed by $\alpha(F) < 18$;
5. omitting uninstalled C_S clauses makes the CNF weaker, so an UNSAT result remains sufficient;
6. the budget-five refutations force any putative budget-six candidate to have exactly six deletions;
7. after choosing one absent f_i , exactly five residual deletion literals are true; and
8. checked UNSAT of all three exact-six branches contradicts the model provided by any candidate graph;
9. checked UNSAT of each singleton-augmented common relaxation proves the corresponding positive edge consequence; and
10. the checked common SAT model establishes only non-vacuity, while its explicit independent 18-set prevents it from being reported as a repair.

APPENDIX C. FROZEN EXACT-SEVEN FIRST-ROUND RECORD

All three endpoints are UNKNOWN. The branch records are content-addressed as follows.

Table 6: Exact-seven bounded-search records.

branch	result JSON SHA-256	checkpoint SHA-256
0	97e777da32b82a83b8e999c9af1f92 7c87b67689b8fb72f8b31ded07b39a 2f7f	918dc1cc80860d97b3209f6d969a15 6b9df5fb8b4df7f0fa910e72fdc577 8098
1	81734a23298ea9c3364a047c3b7b10 910bcd44faf9ff55c3a251fd709f6a c64f	8491cbd5e073170b6f7a503efb3d6a f6c077b6d72defbf9cd6feb6f06e85 fc7f
2	92e2c948892d6f298e0ca3a5055cf3 1cf3eea361b6eee8197ba536922e43 7554	9e8ea23904378ec1bfc6e4c486fd32 f10ccc99c1646855c7589c21cc3c9b 9f5c

The frozen-state checker, its unit tests, and the branch generator have respective SHA-256 identities

checker: 8c1f2629193de8b16e018f036ce462b8e18110b9f4f1c13b3162b25fe415d6f4

tests: 46b0324d67baaf66bae49027bf4ddbfe3aa08684d3cbe364326bc4400914c987

generator: 1 3 a 7 c f e b 5 6 4 2 9 7 f c a d 2 e a 2 c 6 c a 8 3 1 a 6 b
f4b4755e0223abcf9d1bd1178103d0de

Each branch loaded the common universal bank in under six seconds and then spent approximately 300 seconds in its first solver call. The registered endpoints were

branch	fixed absent	solver	endpoint
0	(97, 98)	CaDiCaL	UNKNOWN
1	(97, 99)	Glucose 4.2	UNKNOWN
2	(98, 99)	MapleChrono	UNKNOWN

The state checker verifies formula dimensions, the universal bank digest, the exact counter truth table, exclusion of original nonedges from the counter, and the three timeout records. It does not transform those timeout records into a satisfiability or unsatisfiability claim.

C.1. Deletion-first branch-1 follow-up. The historical ascending implementation used for the first nine deletion-first endpoints has SHA-256 identities

runner: 36dc3b53941605bc4ec132b70b4f61c5afbbcdad13742fe9056a38ddb1683e5a0

tests: ed62d2318e64d6731bca343e829f2e65c058f8f484a72b027b2a87da9f2fd76f

The order-aware implementation used for the subsequent seven gates has runner and test identities

runner: 22ddb00aa93727aa51d2edd6d09b0e536a230e0cc17426004858e98620d6c4f8

tests: b65ebc6f265de6552cdf6f8c09586f4c7c2ae01db0292908ae7998beea75a35c

The sanitized summary identity for all sixteen endpoints is updated in the project-relative machine record named below. The runner’s structural fingerprint is e 9 1 f 0 7 d b 0 f 0 1 b 4 4 2 6 3 f a 7 9 0 9 2 2 e 4 6 e f b a149b2f461d7afb77b53f847b9641839. The project-relative machine summary and human-readable command ledger are `routes/finite/r3_18_budget7_benders_pilot_summary.json` and `routes/finite/R3_18_BUDGET7_BENDERS_PILOT_2026-08-30.md`.

Table 7: Historical ascending branch-1 deletion-first endpoints.

run	endpoint	sec.	cuts	unproved NG
g42_c0	subproblem	3.680	0	0
	oracle limit			
g42_c4096	subproblem	9.700	16	0
	oracle limit			
m22_c0	subproblem	15.256	2816	0
	oracle limit			
m22_c4096	subproblem	47.856	1039	0
	oracle limit			
m22_c16384	global wall	180.026	0	0
maple_c4096	subproblem	62.784	768	2
	oracle limit			
g42+maple	subproblem	121.884	10780	0
	oracle limit			
maple+maple	subproblem	197.868	13362	2
	oracle limit			
m22+maple	global wall	300.095	12800	0

Table 8: Order-aware branch-1 gates.

run	endpoint	sec.	models	cuts
g42_bidir60	global wall	64.077	1	512
maple_bidir60	global wall	60.165	0	0
g42_bidir10	global wall	120.218	5	2560
m22_bidir10	global wall	120.038	1	512
maple_bidir10	global wall	120.059	4	2048
g42_batch4096	global wall	180.200	4	16384
maple_batch4096	global wall	180.089	3	12288

In Table 7, the final column counts only solver-trusted fixed-deletion UNSAT endpoints retained by the search. No DRAT or LRAT proof was emitted

for either endpoint, and no ordered gate added a new no-good. All rows in Tables 7 and 8 therefore have global status UNKNOWN, and none is used in the proof of the finite theorem.

On the two reconstructed deep supports, reverse vertex order returned 512 validated independent 18-sets in 11,261 and 10,057 nodes, about 0.006 seconds each; the ascending order had reached its resource limit. Each nonempty ordered gate likewise returned its requested cut batch in at most 0.075 seconds, but none reached a fixed-deletion subproblem. The same oracle-cap and order/batch/master matrix was therefore stopped rather than extrapolated.

C.2. Proof-carrying singleton consequences. The follow-up common relaxation Ψ_1 has 154190 variables and 718452 clauses before candidate units. Its formula fingerprint is `a6f920afa451174ed05932174951481a63647a2ba603c762379f9357323fa5e2`. It contains 161700 triangle clauses, 19680 exact-six counter clauses, one fixed branch unit, 285300 degree-cap clauses, and the ordered bank of 251771 universal independent-18 clauses. No learned core clause is installed. Appending one candidate-deletion unit gives 718453 clauses.

The proof-run exporter is frozen by Git commit `408e3bb` and SHA-256 `5e05f75affc5c6c5323fb2f1bddd3097c7a851c00330fd558f1f731e978936cf`. The final four-singleton machine summary has SHA-256 `8ae4f0d9ea919c915ba642eb03ec3d418b4419a89781377509acbe925d89555a`. The strongest released antichain consists of the following four singleton refutations; the earlier pair refutations are valid but dominated and are not included as normative release assets.

Table 9: Branch-1 singleton-proof payloads.

label	assumed deleted edge	CNF bytes	DRAT bytes
K_a	(11, 62)	18914252	707650273
K_b	(18, 61)	18914251	855530402
K_c	(18, 64)	18914252	1117799611
K_d	(18, 69)	18914248	811556002

The corresponding compressed CNF identities are, in table order,

K_a: `1ec7a15b9afd7c7c0944bbc663db2522b47d6f4f12fe98f23501d1006fabe411`
K_b: `1df74150cf9678e977364d4df782e179d62dff811b940d9fc0ea35c30cda845b`
K_c: `95efbf95802cf05e7ce5d676c35470989b126b6145eb00f7e67a546948bde07e`
K_d: `b852ca92b33ac461d425f62309486e30645f36f7736bd18b2b5e15b90fb27483`

The compressed DRAT identities are, in the same order,

K_a: `c71be4804965a66e2783c61ae4711e1dc5faeb13a42ecab50672dd0efdd8818c`
K_b: `c34b97f66e3513c8261e1424f66b9e2299a49b9f38a6d953ade17e6c4e266bfa`
K_c: `375abf0884e415c2f4fea15d1971b818b2535996f017269fee742e90376f7fc3`
K_d: `34853e20f7055e9a1e33b2668437046414e37a06821a82526eb4c822e61ab886`

CaDiCaL was pinned to source commit `c60730422e758ef1cebe7aeddff2dda31c996bf04`; `drat-trim` was pinned to `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. Each solver run exited 20 with a complete proof, and each checker run exited zero with an exact `s VERIFIED` line. A separately compiled arm64 checker from the same source commit performed a separate cross-architecture replay of all four traces. This is a second binary replay, not a second implementation of DRAT semantics.

The PySAT-free semantic checker has SHA-256 `2dcc16137a0bd555a2553e9987968fda429e9098ec8c21affb05bfd5a5e03f3b`. It parses the seed and ordered bank, reconstructs both counter families without importing the production builder, and compares the complete DIMACS byte stream. Its four audit-record SHA-256 identities are

```
K_a: 4637a558f46abefcfafcf5940781eb4c9c76556793491e48b1eb5e919d9d6d581
K_b: c5d1f6823fb08388517c4eb9e0cddc040e27d080168ff4bd513928e095a908a9
K_c: 3a95e9e412d78c2fb7af1e1ff68828846958a5f969a70996051d2c5dcbbef9bf
K_d: f38e987e9fecdd7c4ed646cd9c0f50a3838f8ee6cadd436cd4b563d5db431b26c
```

The common relaxation itself is nonempty. Its deterministic compressed CNF has SHA-256 `39249ef8378de3f2ef412e514f6283cdaa032bc17fb25fc3448463e5566f5365`; the complete compressed model has SHA-256 `9057db25b785640345e6f724cb1c79313642bb62ec2a19d9f310e25700de024d`. An independent audit with SHA-256 `7b2af8d92d35721fd6197b3ac3c5d52066bd04faa545e219cc57b94dce035921` evaluated all clauses and recovered the independent 18-set displayed in the main text. Thus this witness proves non-vacuity but is not a target repair.

The two historical fixed-deletion supports in Table 7 delete both (11, 62) and (18, 61). Each is therefore excluded already by either of the corresponding singleton certificates. Their original solver calls emitted no proofs and are not retroactively described as certified; the later common-relaxation proofs are a distinct evidence chain.

The machine-readable proof record is `routes/finite/r3_18_budget7_branch1_core_proof_summary.json`; the full human ledger is `routes/finite/R3_18_BUDGET7_BRANCH1_CORE_PROOFS_2026-08-30.md`. The four singleton CNF/DRAT pairs, the common CNF/model pair, and two audited checker binaries are designated large release assets rather than Git objects. Their public-release download and clean-clone replay remain a separate publication gate and do not alter the exact-seven status UNKNOWN.

C.3. Maximal-union construction and bounded endpoint. A standalone degree-initialized one-dimensional Weisfeiler–Leman checker certifies the color-class profile (3,15,100), and hence the triviality of the seed automorphism group. Its script, three-test regression, and audit-record SHA-256 identities are `613c902fe89fedfc2e2132e6a5d7ea182b7cc4b2fd245885861c29f36ba62f18`, `93047e1673d73f6adca0109bb200d524191b9f9671d0ed84b7466203f66cc114`, and `8ad9fbec731a3c2f3b3012695022366fc66070265c6f5a27bfc0c66f34e872ce`. Thus the branch-1 gate below has no seed-symmetry implication for branches zero and two.

The direct branch-1 maximal-union formula combines the four checked retention units, all 530525 distinct independent-18 masks in the four-family union, and the full 4950-pair maximal-triangle-free selector encoding. Its identity is

variables	639290
clauses	1972360
bytes	408370088
SHA-256	09e1784c3f43c4901dc6f6b4749fc5a74b025b08f74be58133edd1ed1096ebdb

The production formula record and independent reconstruction audit have SHA-256 identities `cb3eaa1ed462713e13a951dbc113555ce8f1aa8325c266658d9cc32c8dd72026` and `551e9188a4b7eb42bb40080acb0b2378ca6af69d412fb51f0c964ec2460190d2`, respectively. A post-review rebuild retained these two record identities and the exact raw-CNF identity above.

The final generator, independent formula checker, and their 11-test regression have respective SHA-256 identities

generator: `c07c76c5c6dde10474c10d48703270b83cff5a8865d223e7674342227dbafaf`
4

checker: `9458ae9cdccc380efc1d083d6a6c06c14bd7a49ac728ca43c836ee5b62bacda2`

tests: `2a85014c95de3e13e642ee91218cada552f4b4419c04c3e73a8b530465418df1`

The tests include command-line regressions proving that authenticated input symlinks are not dereferenced before the non-symlink check.

One Sirius invocation used CaDiCaL 3.0.1 from source commit `c60730422e758ef1cebe7aeddff2dda31c996bf04`; the executable SHA-256 was `bd054bcc5864fd20c9ff117f8fff94f810f5b210014ff33971a6c1db1d0eca45`. The external wall was 300 seconds and the wrapper returned 124; the solver result was exactly `c UNKNOWN`. The incomplete DRAT prefix had 922038097 bytes and SHA-256 `a6693fe2b3bab57f9d0e9ebe39bd7baa3883bfe2af15a5eb33b0aded972b5438`. It was not replayed and was deleted after hashing. In particular, it is not a partial certificate and supplies no evidence toward unsatisfiability.

The sanitized endpoint record and its fail-closed audit have SHA-256 identities `e35f7761f92ddebdb4029553fe7a2828e31c79d68820d37b1aa2bb6ac4f80571` and `4e4e212f0c6bae101db00dfb118f602cc6f874e9d8833f3c43cc946432e2cc54`. The endpoint checker and its three-test regression have identities `7c6879bcd9c07ed27ebf25cb601a4ce8c69e5714178e35a9fb286c740b3c2ad0` and `83f545543792e22d9d825ae22ee9b00883172cf902a21d8b7be368cc80e736bd`. They enforce the status `MAXIMAL_UNION_TIMEOUT_UNKNOWN_NO_CUT`; no branch or global conclusion is attached to this endpoint.

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